

Hwanseok Chang

Ph.D. Candidate | Materials Science and Engineering | Korea University

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Research Profile

My research focuses on inverse design and scalable manufacturing of functional 3D nanoarchitectures. I develop algorithms that determine optical fields and photomask geometries for large-area fabrication of prescribed structures, and extend this framework toward performance-driven discovery of architectures for sensing, structural, thermoelectric, photonic, and other metamaterial functions. Experimentally, I fabricate 3D polymer templates and convert them into ceramic shells, metals, and glassy carbon while preserving complex geometry. By integrating inverse design, scalable nanofabrication, material conversion, and AI-assisted optimization, I aim to create architected metamaterials that overcome intrinsic material limitations through structure.

Education

2025.03 – Present	Ph.D. in Materials Science & Engineering, Korea University, Seoul
2023.03 – 2025.02	M.S. in Materials Science & Engineering, Korea University, Seoul Thesis: Inch-scale fabrication of 3D thin-shell alumina with woodpile structure and their application in lightweight and high-strength structural material
2017.03 – 2023.02	B.S. in Materials Science & Engineering, Korea University, Seoul
2014.03 – 2017.02	Hansung Science High School, Seoul

Experience & Activities

2025.03 – Present	Ph.D. Candidate, Nanomaterials for Electronics and Optics Lab., Korea University
2025.01 – 2025.12	Lab Manager, Nanomaterials for Electronics and Optics Lab. (NEOLab), Korea University
2023.03 – 2025.04	Research Staff, MISODI Project, Korea University Development and application of optimal 3D nano-modular materials by Fourier inverse design
2022.09 – 2022.12	Research Intern, Electronic Thin Films Lab., Korea Institute of Science and Technology (KIST)
2025.04	Poster Presenter, 2025 Spring Meeting of the Korean Ceramic Society
2024.10	Poster Presenter, 2024 Fall Meeting of the Korean Ceramic Society
2024.07	Staff, 2024 BrainLink X-Lab Day
2023.10 – 11	Poster Presenter, 13th A3 Symposium on Emerging Materials

Honors & Scholarships

2026	BK21 International Joint Training Scholarship for Outstanding Graduate Students, Korea University
2026	Doctoral Excellence Scholarship (Science & Engineering), National Research Foundation of Korea
2025.10.17	Changsung Outstanding Poster Award, 2025 Fall Meeting of the Korean Ceramic Society “Inch-Scale Fabrication of 3D Thin-Shell Alumina with Diamond Structure and Their Application in Lightweight and High-Strength Structural Material”
2025.10.15	Korea Ceramic Federation President’s Award, 2025 Ceramics Day

2025.04.18	KCerS Excellence Poster Award, 2025 Spring Meeting of the Korean Ceramic Society
	“Photolithographic Realization of Target Nanostructures in 3D Space through Optical Modulation”
2021.2	Top Student of the Semester, Korea University
2021.1	Outstanding Student of the Semester, Korea University
2017.2	Outstanding Student of the Semester, Korea University
2016	Ministerial Award, National Science Exhibition

Publications

Published/accepted: 3 first/co-first-author + 3 co-authored · In preparation: 2 first/co-first-author manuscripts

† Equal contribution; * Corresponding author

1. **Hwanseok Chang**†, Sungjoo Kwon†, Gwangmin Bae*, and Seokwoo Jeon* *Rational design of arbitrary topology in three-dimensional space via inverse calculation of phase modulation*. Nanophotonics, 13(7), 971–982 (2024).
2. Jewook Kim†, **Hwanseok Chang**†, Gwangmin Bae, Myungwoo Choi*, and Seokwoo Jeon* *Graphene-based thermoelectric materials: toward sustainable energy-harvesting systems*. Chemical Communications, 61(27), 5050–5063 (2025).
3. Junhyung Park†, Yun Hyeong†, **Hwanseok Chang**, Dohyun Kim, Joonkyo Jung, Seokwoo Jeon, and Jonghwa Shin* *Spectrally Encoded Proximity-Field Nanopatterning*. Advanced Optical Materials, e03274 (2026).
4. Juheon Lee, Yeon Wook Kim, **Hwanseok Chang**, Herve Hugonnet, Chulmin Oh, Sehyeon Lee, Seung-Mo Hong, Seokwoo Jeon, and YongKeun Park* *Incoherent dielectric tensor tomography for quantitative three-dimensional measurement of biaxial anisotropy*. Nature Photonics, 1–10 (2026).
5. Gye Hyeon Lee†, Youngoh Kim†, Junhwa Oh†, Yun-Haeng Cho†, Sunwoo Lee, Jewook Kim, Minhyeok An, Dong Gyun Hong, **Hwanseok Chang**, Seung Yol Jeong, Matthew T. Flavin, Jeong-O Lee, JungHoon Lee, Jun Min Suh, Seokwoo Jeon, Young-Seok Shim*, Seyong Oh*, Myungwoo Choi*, and Donghwi Cho* *Photoactivated Chemical Sensing Payload for Drone-Enabled Remote Environmental Monitoring*. Nature Communications (accepted, 2026).
6. **Hwanseok Chang**†, Sunwoo Lee†, Youngkwon Kim†, Yea-Lee Lee, Yun-Gyu Lee, Jae Hun Hwang, Gyu-Li Kim, Gye Hyeon Lee, Dong Gyun Hong, Seung Yol Jeong, Yongseong Kim, Hak-Young Ahn, Seonggwang Yoo, Young-Seok Shim, Jung-Hoon Lee, Taeyong Eom, Jeong-O Lee, Seokwoo Jeon, Kwangjae Lee*, Myungwoo Choi*, and Donghwi Cho* *Activation-Free Gas Sensing via Physisorption-Engineered 3D van der Waals Nanoarchitectures*. Nature Communications (accepted, 2026).
7. Gwangmin Bae^{1,†}, **Hwanseok Chang**^{1,†}, Jihoon Jeong^{2,†}, Tae Hee Lee³, Samuel Hwang¹, Sang-Hyeon Nam⁴, Jung-Wuk Hong^{3,*}, Dongchan Jang^{2,*}, and Seokwoo Jeon^{1,*} *Scalable Diamond-TPMS Ceramic Nanoarchitecture with Shear-Band-Mediated Damage Tolerance*. (In preparation).
8. Junhyung Park†, **Hwanseok Chang**†, Myungjoon Kim, Nayoung Kim, Gwangmin Bae, Jonghwa Shin*, and Seokwoo Jeon* *Prescribed volumetric light-field synthesis for single-exposure 3D nanolithography*. (In preparation).

Patents

1. Jonghwa Shin, Junhyung Park, Myungjoon Kim, Nayoung Kim, Seokwoo Jeon, Gwangmin Bae, and **Hwanseok Chang**. 포토마스크 제조방법 및 이를 기반으로 하는 자이로이드 나노구조의 제조방법. Korean Patent Application 10-2025-0148095.
2. Jonghwa Shin, Junhyung Park, Myungjoon Kim, Nayoung Kim, Seokwoo Jeon, Gwangmin Bae, and **Hwanseok Chang**. 포토마스크 제조방법 및 이를 기반으로 하는 다이아몬드 또는 장작더미 나노구조의 제조방법. Korean Patent Application 10-2025-0148102.
3. Seokwoo Jeon, Dongchan Jang, Gwangmin Bae, **Hwanseok Chang**, and Jihoon Jeong. 위상 제어 2 단계 근접장 나노패터닝을 이용한 대면적 다이아몬드형 삼중주기 최소곡면 나노구조체, 이를 포함하는 경량·고강도 구조재료 및 그 제조방법. Korean Patent Application planned (Management No. DP 2026-0900).

Research Interests

Inverse design and AI-assisted optimization of 3D nanoarchitectures; scalable 3D nanofabrication; architected materials and metamaterials for sensing, structural, photonic applications; material conversion

Technical Skills

Simulation, AI & Programming: FDTD (Lumerical), RCWA (Lumerical), Python, MATLAB, inverse optimization/PSO, AI-assisted scientific programming, machine-learning model training, data analysis and automation
Nanofabrication: E-beam lithography, 3D/proximity-field lithography, photomask processing, mask aligner, nanoimprint/PER processing, ALD, e-beam evaporation, FIB, ellipsometry
Characterization: SEM, TEM, XRD, XPS, Raman, GIWAXS